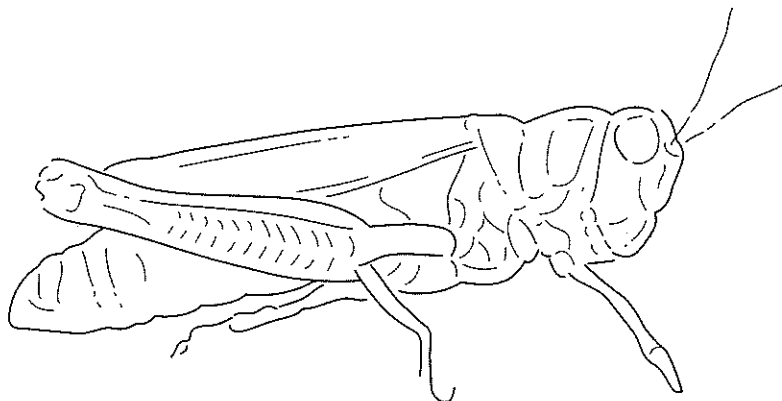




All living things are suited to their living conditions. The features which enable the organisms to live in specific conditions are called *adaptations*. If an organism is not adapted to its environment, it will die there. Adaptations can be behavioural, physiological and physical (= structural). Below is a diagram of a grasshopper and some of its adaptations.



Behavioural adaptations

How does it behave? A grasshopper chooses to quiver in order to mimic the movement of leaves.

Physical adaptations

What is it like? The colour of the grasshopper camouflages it from its predators and its strong hind legs allow it to jump great distances.

Physiological adaptations

How is the body reacting? The grasshopper lays many eggs and is capable of digesting large amounts of plant leaves and shoots.

- 1** Read the information below about the cane toad. Classify its adaptations as behavioural, physiological or physical.

The toad was introduced to Australia to get rid of a pest. It was well adapted to the harsh Australian climate. The toad has rough, brown, dull coloured skin giving it a perfect camouflage against predators. The toad's skin is also very tough with poisonous glands. This deters many predators. Toads also have the capability of puffing up their bodies to make them look larger and more frightening. This scares many predators away. Female cane toads lay large numbers of eggs. They can lay them in fresh or salty water. Toads have been seen to eat a range of insects, and even canned pet food.

- a Behavioural adaptations: _____

- b Physiological adaptations: _____

- c Physical adaptations: _____

- 2** Humans also have many adaptations which make it possible for them to live in many different habitats. For the habitats described below, list in the following table several human adaptations. For each adaptation state whether it is behavioural, physiological or structural and explain the purpose of that adaptation.

- a a person living in snow country
 b a person on the beach in summer
 c a person working outdoors in central Australia

To be adapted or not to be adapted 18b



Situation	Adaptation	Category	Purpose

Looking at language

Cause and effect conjunctions include such words as *because, since, therefore, so, as a result* and *consequently*. These conjunctions can be used to join short sentences into more complex sentences.

- 3** Complete the table below using cause and effect conjunctions. After constructing each sentence classify each adaptation as behavioural, physiological or physical.

Short sentences	Long sentence	Type of adaptation
a Elephants have large ears. They can cool off quickly.	Elephants have large ears and as a result they can cool off quickly.	physical
b Kangaroos lick their paws. Their spit evaporates. Their paws become cooler.		
c Lizards move to the shade on hot days. They stay cool.		
d Eagles have sharp, curved beaks. They can tear meat.		
e Rabbits rest in their burrows. They stay cool.		
f Dolphins have a streamlined body. They can swim fast.		
g Birds have light bones and wings. They can fly.		
h Kookaburras have a tough beak. They can eat snakes.		
i The cactus has spines for leaves. Spines have no pores. Water is conserved.		
j Dogs shiver on cold mornings. They get warm.		
k Koalas sleep in the middle of the day. They don't get too hot.		
l Frogs have slimy moist skin. They can breathe through their skin.		